



SECURITY ASSESSMENT Presale Escrow

September 7, 2026

Audit Status: Pass

BLADE POOL

RISK ANALYSIS | Presale.

■ Classifications of Manual Risk Results

Note: Risk classifications follow CVSS v4.0 standards (<https://www.first.org/cvss/v4.0/specification-document>). Contract security metrics are verified through GoPlus API token scanner technology, which provides real-time validation of contract parameters including

Classification	CVSS Range	Description
 Critical	9.0 - 10.0	Critical vulnerabilities that pose immediate and severe risks requiring urgent attention.
 High	7.0 - 8.9	High-priority issues that could lead to significant security breaches or financial loss.
 Medium	4.0 - 6.9	Medium-severity findings that should be addressed to improve contract security.
 Low	0.1 - 3.9	Low-risk items or best practice suggestions with minimal security impact.
 Informational	0.0	Informational findings about detected functions or contract features.

■ Manual Code Review Risk Results

Contract Security	Description
 Buy Tax	N/A
 Sale Tax	N/A
 Cannot Buy	N/A
 Cannot Sale	N/A
 Max Tax	N/A
 Modify Tax	N/A
 Fee Check	Not Applicable
 Is Honeypot?	Not Applicable
 Trading Cooldown	Not Detected
 Enable Trade?	N/A

Contract Security	Description
● Pause Transfer?	Not Detected
● Max Tx?	Not Detected
● Is Anti Whale?	Not Detected
● Is Anti Bot?	Not Detected
● Is Blacklist?	Not Detected
● Blacklist Check	Not Detected
● i Is Whitelist?	Not Detected
● Can Mint?	N/A
● Is Proxy?	Not Detected
● Can Take Ownership?	Owner receives 85% on finalize; liquidity provisioning (graduate) is owner-discretionary
● Hidden Owner?	Not Detected
● Owner	Sale owner (creator) - controls finalize payout, graduate, abort, whitelist
● Self Destruct?	Not Detected
● External Call?	Present (router / trusted wiring)
● i Other?	Contribution escrow, softcap/hardcap, early-withdraw fee, finalize (85% to owner / 15% LP), graduate (LP burned to DEAD), abort/refund
● i Holders	N/A
● Audit Confidence	Good - Pass; elevated centralization / rug risk noted (owner controls 85% of raise + discretionary liquidity)
● Authority Check	Owner-controlled
● Freeze Check	N/A



Presale

Executive Summary

TYPES

DeFi

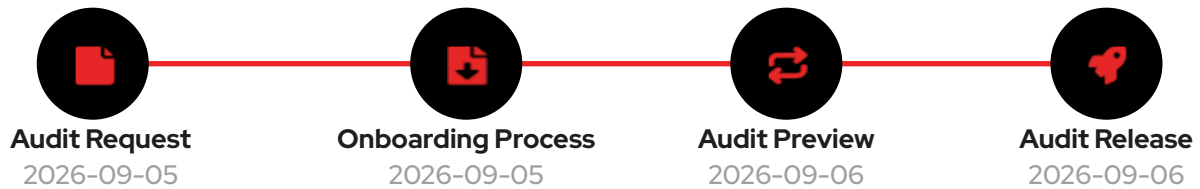
ECOSYSTEM

Robinhood Chain

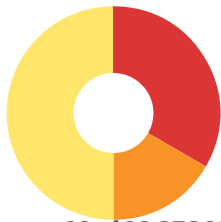
LANGUAGE

Solidity

Timeline



Vulnerability Summary



6
Total Findings

0
Resolved

6
Pending

6
Unresolved

20 of 26 CFG01-CFG26 checks passed with no security finding; 6 open findings. Slither 0.11.5 static analysis.

Severity	Resolution Status	Description
0 Critical		Immediate danger. Can lead to loss of user funds, unauthorized access, or full system compromise.
2 High	0 Resolved, 2 Pending	Significant risk that should be addressed urgently to prevent exploitation or financial loss.
1 Medium	0 Resolved, 1 Pending	Moderate risk that can lead to issues if combined with other weaknesses; should be addressed.
3 Low	0 Resolved, 3 Pending	Minor concern or best-practice deviation with limited direct security impact.
0 Informational		Not a vulnerability - code-quality, documentation, or optimization suggestions.

Industry Standards Compliance

This audit follows internationally recognized security standards to provide comprehensive assurance.

EEA EthTrust Security Levels

EthTrust/EVM review conducted for a Robinhood Chain contract.

Smart Contract Security Verification Standard (SCSVS)

SCSVS review conducted for an EVM smart contract on Robinhood Chain.

CVSS v3.1 Severity Scoring

Per-finding CVSS recorded on each CFG item.

PROJECT OVERVIEW | Presale.

Token Summary

Parameter	Result
Address	0x9fBb80283901B25E6BDacC0698a9Be4E0a48722c
Name	Presale
Token Tracker	Presale (HOODPAD)
Decimals	N/A
Supply	N/A (escrow)
Platform	Robinhood Chain
Compiler	v0.8.34+commit.80d5c536
Contract Name	Presale
Optimization	Yes with 200 runs
LicenseType	MIT (SPDX)
Language	Solidity
Codebase	https://robinhoodchain.blockscout.com/address/0x9fBb80283901B25E6BDacC0698a9Be4E0a48722c?tab=contract

■ Main Contract Assessed

Name	Contract	Live
Presale	0x9fBb80283901B25E6BDacC0698a9Be4EOa48722c	Yes
PresaleFactory (deployer)	0x2ce5caa81abe7af049a079dd2bcefd42ea7b85a2	Yes
TaxToken (sale token)	0xBC43946C88e44c4f0F84d26FDfB1a9aa76BcC592	Yes

■ TestNet Contract Was Not Assessed

■ Solidity Code Provided

SolID	File Sha-1	FileName
Presale	d37884015ff408286f5d3fd91f9fb686e8028184	HOODPAD.sol

TECHNICAL FINDINGS

Smart contract security audits classify risks into several categories: Critical, High, Medium, Low, and Informational. These classifications help assess the severity and potential impact of vulnerabilities found in smart contracts.

Classification of Risk

Severity	CVSS Range	Description
 Critical	9.0 - 10.0	Immediate danger. Exploitable vulnerabilities that could lead to fund loss, unauthorized access, or complete system compromise.
 High	7.0 - 8.9	Significant security risk. Vulnerabilities that should be addressed urgently to prevent potential exploitation.
 Medium	4.0 - 6.9	Moderate security risk. Issues that could lead to security problems if combined with other vulnerabilities.
 Low	0.1 - 3.9	Minor security concern. Best practice violations or low-impact issues.
 Informational	0.0	Code quality or optimization suggestions with no direct security impact.

Slither Static Analysis Results

Slither is a Solidity static analysis framework that runs a suite of vulnerability detectors to identify potential security issues, code quality problems, and best practice violations. It provides comprehensive security analysis with confidence ratings for each finding. For full detector documentation, visit: <https://github.com/crytic/slither/wiki/Detector-Documentation>

Impact Level	Count	Description
High	N/A	Critical security vulnerabilities requiring immediate attention (e.g., reentrancy, unchecked transfers)
Medium	N/A	Significant issues that should be addressed (e.g., dangerous comparisons, arithmetic issues)
Low	N/A	Minor issues and best practice violations (e.g., missing events, naming conventions)
Informational	N/A	Code quality and optimization suggestions (e.g., unused variables, gas optimizations)
TOTAL	N/A	Total findings across all severity levels

Analysis Summary

Automated Slither analysis was not run for this build (solc 0.8.34 exceeds the host's pinned solc-select versions). The review is a manual line-by-line analysis of the verified flattened Blockscout source against CFG01-CFG26. PRELIMINARY report for client review.

Remediation Approach

All High and Medium impact findings from Slither have been manually reviewed and mapped to CFG test cases where applicable. Each finding has been analyzed in the context of the contract's business logic to determine actual exploitability. Detailed analysis of specific findings is available in the Technical Findings section of this audit report.

Deprecation Notice

The Smart Contract Weakness Classification Registry (SWC Registry) was deprecated in September 2023. CFGNinja audits now use Slither for static analysis, which provides more comprehensive and up-to-date vulnerability detection. SWC was an implementation of the weakness classification scheme proposed in EIP-1470, loosely aligned to the Common Weakness Enumeration (CWE). For historical reference: <https://swcregistry.io/>

Detailed Security Findings

Each finding lists its severity, CVSS, location and description, the affected source excerpt (verbatim from the verified Blockscout source), our recommendation, mitigation, references, and a suggested fix where the remediation is code-expressible.

HOODPAD-04 | abort() After finalize() Can Strand Buyer Refunds..

Category	Severity	CVSS	Location	Status
Centralization / Denial-of-Service Vectors	 High	7.5	HOODPAD.sol: Presale.finalize() / abort() / refund()	 Detected

Description

finalize() (callable by anyone once the sale ends) pays the owner 85% of the raise (toOwner = address(this).balance - liquidityEth) and keeps only ~15%. The owner can then call abort() (which only requires !liquidityTaken), setting softCapReached = false and burning the held tokens. Buyers now fall into the refund() path (which requires finalized && !softCapReached), but the contract holds only ~15% of the raise - so refunds drain that and then revert, and claim() also reverts (softCap false). Buyers of a SUCCESSFUL sale can lose ~85% of their contribution.

Recommendation

Disallow abort() once finalize() has released the owner payout (e.g. block abort when finalized && softCapReached), or escrow the owner payout until liquidity is added / the refund window closes. Make the finalize and abort state transitions mutually exclusive.

Mitigation

finalize()->abort() sequencing lets a successful-sale owner strand buyer refunds after taking 85%.

References:

<https://swcregistry.io/docs/SWC-105>

<https://owasp.org/www-project-smart-contract-top-10/>

HOODPAD-08 | Duplicated finalize() Logic in graduate()..

Category	Severity	CVSS	Location	Status
Code Quality	 Low	1.5	HOODPAD.sol: Presale.graduate()	 Detected

Description

graduate() re-implements the success branch of finalize() (the caps check, owner payout and burn) rather than calling a shared internal routine. Duplicated logic can diverge during maintenance and makes the state machine harder to reason about.

Recommendation

Extract the shared finalize logic into a single internal function called by both finalize() and graduate().

Mitigation

finalize() success logic is duplicated inside graduate().

References:

<https://consensys.github.io/smart-contract-best-practices/>

HOODPAD-10 | Systemic 15% Liquidity Ratio..

Category	Severity	CVSS	Location	Status
Liquidity Security	 Medium	5.0	HOODPAD.sol: Presale. LIQUIDITY_BPS (1500)	 Detected

Description

LIQUIDITY_BPS = 1500 is a hardcoded platform default applied to EVERY sale on the pad: 85% of the raise goes to the owner and only 15% to listing liquidity. Buyers cannot choose a higher lock ratio per sale. Combined with the discretionary graduate() (CFG20), this is a high rug-risk ratio that contributors should understand before participating.

Recommendation

Make the liquidity ratio a per-sale, buyer-visible parameter with a sane minimum, or raise the default. Surface the ratio prominently to contributors in the launchpad UI.

Mitigation

Fixed 15% liquidity / 85% owner split on every sale; not buyer-configurable.

References:

<https://swcregistry.io/docs/SWC-101>

<https://swcregistry.io/docs/SWC-105>

<https://owasp.org/www-project-smart-contract-top-10/>

HOODPAD-18 | 20% Early-Withdraw Fee..

Category	Severity	CVSS	Location	Status
Transparency	 Low	2.0	HOODPAD.sol: Presale. EARLY_FEE_BPS (2000) / withdrawEarly()	 Detected

Description

EARLY_FEE_BPS = 2000 charges a 20% fee to contributors who exit early via `withdrawEarly()`; the fee is paid to the launchpad. This is a disclosed protocol constant rather than a hidden mechanism, but the size of the penalty should be made obvious to contributors.

Recommendation

Ensure the 20% early-exit penalty is clearly communicated to contributors in the launchpad UI before they commit.

Mitigation

Disclosed 20% early-withdraw penalty routed to the launchpad.

References:

<https://consensys.github.io/smart-contract-best-practices/>

HOODPAD-20 | Liquidity Is Not Guaranteed; 85% Released to Owner Up Front..

Category	Severity	CVSS	Location	Status
Centralization / Ownership	 High	7.1	HOODPAD.sol: Presale.finalize() / graduate()	 Detected

Description

On a successful sale, finalize() immediately sends 85% of the raise to the owner with no lock or vesting. Liquidity is added ONLY if the owner voluntarily calls graduate() (onlyOwner); nothing in the contract forces it. An owner can take the raise and never provide liquidity, leaving buyers holding claimable-but-illiquid tokens. When graduate() is called, the LP is burned to the dead address (permanently locked - a positive), but it represents only the 15% reserve.

Recommendation

Provision liquidity atomically within finalize() (so it cannot be skipped), and/or escrow or vest the owner payout behind successful liquidity provisioning. Remove owner discretion over whether liquidity is ever created.

Mitigation

Owner receives 85% immediately; liquidity depends entirely on a discretionary graduate() call.

References:

<https://swcregistry.io/docs/SWC-101>

<https://swcregistry.io/docs/SWC-105>

<https://owasp.org/www-project-smart-contract-top-10/>

HOODPAD-23 | Anonymous Team - No KYC Verification..

Category	Severity	CVSS	Location	Status
Team Transparency	 Informational	0.0	HOODPAD.sol: Project team	 Detected

Description

No team member names, credentials or verifiable identities are disclosed, and no KYC has been completed with a recognized provider. This is flagged for transparency and is the reason the audit score is capped at 90 under CFG Ninja policy (a higher score requires KYC). It does not, by itself, indicate a code vulnerability.

Recommendation

Complete KYC with a recognized provider (e.g. Assure DeFi, PinkKYC, SolidProof) and publish at least a team-lead identity to lift the score cap.

Mitigation

No KYC or team disclosure found; score capped at 90.

References:

N/A

HOODPAD-24 | Standalone Deployment Trusts an Arbitrary Token..

Category	Severity	CVSS	Location	Status
Input Validation	 Low	2.0	HOODPAD.sol: Presale.constructor (Cfg.token)	 Detected

Description

The Presale constructor accepts an arbitrary token address. Within the factory flow the token is always a freshly deployed TaxToken, but a Presale deployed directly (outside the factory) could be pointed at a malicious token. This is informational for the pad's intended flow but worth noting for anyone integrating the contract standalone.

Recommendation

For the launchpad, only use presales created by the audited factory; optionally validate the token against the factory registry inside the constructor.

Mitigation

Constructor accepts any token; only the factory-created path is guaranteed safe.

References:

<https://swcregistry.io/docs/SWC-105>

<https://owasp.org/www-project-smart-contract-top-10/>

FINDINGS

In this document, we present the findings and results of the smart contract security audit. The identified vulnerabilities, weaknesses, and potential risks are outlined, along with recommendations for mitigating these issues. It is crucial for the team to address these findings promptly to enhance the security and trustworthiness of the smart contract code.

Severity	Found	Pending	Resolved
 Critical	0	0	0
 High	2	2	0
 Medium	1	1	0
 Low	3	3	0
 Informational	0	0	0
Total	6	6	0

In a smart contract, a technical finding summary refers to a compilation of identified issues or vulnerabilities discovered during a security audit. These findings can range from coding errors and logical flaws to potential security risks. It is crucial for the project owner to thoroughly review each identified item and take necessary actions to resolve them. By carefully examining the technical finding summary, the project owner can gain insights into the weaknesses or potential threats present in the smart contract. They should prioritize addressing these issues promptly to mitigate any risks associated with the contract's security. Neglecting to address any identified item in the security audit can expose the smart contract to significant risks. Unresolved vulnerabilities can be exploited by malicious actors, potentially leading to financial losses, data breaches, or other detrimental consequences. To ensure the integrity and security of the smart contract, the project owner should engage in a comprehensive review process. This involves understanding the nature and severity of each identified item, consulting with experts if needed, and implementing appropriate fixes or enhancements. Regularly updating and maintaining the smart contract's codebase is also essential to address any emerging security concerns. By diligently reviewing and resolving all identified items in the technical finding summary, the project owner can significantly reduce the risks associated with the smart contract and enhance its overall security posture.

SOCIAL MEDIA CHECKS | Presale.

Social Media		URL	Result
Website		https://voltipad.com	Pass
Telegram			N/A
Twitter			N/A
Facebook			N/A
Reddit	N/A		N/A
Instagram	N/A		N/A
CoinGecko			N/A
Github	N/A		N/A
CMC			N/A
Email			Contact
Other			N/A

From a security assessment standpoint, inspecting a project's social media presence is essential. It enables the evaluation of the project's reputation, credibility, and trustworthiness within the community. By analyzing the content shared, engagement levels, and the response to any security-related incidents, one can assess the project's commitment to security practices and its ability to handle potential threats.

Social Media Information Notes:

Auditor Notes: Project front-end voltipad.com confirmed (earlier domains hoodpresales.com and voltedhood.com were reputation-flagged and rotated). No CoinGecko/CMC listing at time of audit. Team is anonymous; no KYC.

Project Owner Notes: Public launchpad front-end confirmed active.

Assessment Results**Final Audit Score HOODPAD.**

Review	Score
Security Score	84
Auditor Score	84

Our security assessment or audit score system for the smart contract and project follows a comprehensive evaluation process to ensure the highest level of security. The system assigns a score based on various security parameters and benchmarks, with a passing score set at 80 out of a total attainable score of 100. The assessment process includes a thorough review of the smart contracts codebase, architecture, and design principles. It examines potential vulnerabilities, such as code bugs, logical flaws, and potential attack vectors. The evaluation also considers the adherence to best practices and industry standards for secure coding. Additionally, the system assesses the projects overall security measures, including infrastructure security, data protection, and access controls. It evaluates the implementation of encryption, authentication mechanisms, and secure communication protocols. To achieve a passing score, the smart contract and project must attain a minimum of 80 points out of the total attainable score of 100. This ensures that the system has undergone a rigorous security assessment and meets the required standards for secure operation.



Important Notes for HOODPAD

HOODPAD - Presale - Smart Contract Security Audit Report (PRELIMINARY)

PROJECT OVERVIEW

HOODPAD is a presale launchpad on Robinhood Chain (an Ethereum-EVM L2, chainId 4663). Presale is the per-sale contribution escrow deployed by the HOODPAD factory: it accepts buys within a window, enforces soft/hard caps and per-address limits, and on a successful sale releases 85% of the raise to the owner while reserving 15% for listing liquidity that the owner provisions via graduate() (the LP is burned to the dead address). It also supports early withdrawal (20% fee), abort and refunds. This report covers the Presale contract; the PresaleFactory and TaxToken are covered in separate reports. This is a PRELIMINARY report issued for client review.

The launchpad is trust-minimized on the TOKEN side (fixed tax, ownership renounced) but trust-heavy on the PRESALE side: the owner controls 85% of the raise and whether liquidity is ever provided. That is the honest headline of this report.

CONTRACT DETAILS

Name: Presale

Type: Per-sale contribution escrow

Contract Address: 0x9fBb80283901B25E6BDacC0698a9Be4E0a48722c

Chain: Robinhood Chain (EVM, chainId 4663)

Compiler: Solidity v0.8.34+commit.80d5c536 | License: MIT (SPDX)

Verification: Verified (flattened source published) on Blockscout

Front-end: <https://voltipad.com/>

Explorer: <https://robinhoodchain.blockscout.com/address/0x9fBb80283901B25E6BDacC0698a9Be4E0a48722c?tab=contract>

MECHANICS

- `buy()`: within `[startTime, endTime]`, not finalized, whitelist (if enabled), `msg.value >= minBuy`, cumulative `<= maxBuy`, raised + value `<= hardCap`. Marks `softCapReached` when raised `>= softCap`.
- `withdrawEarly()`: before finalize and before `endTime`; refunds the contribution minus a 20% early-exit fee (`EARLY_FEE_BPS = 2000`) that is sent to the launchpad.
- `finalize()`: callable by anyone once ended or hardcap reached. On success (raised `>= softCap`) it sends 85% of the raise to the owner and keeps 15% (liquidityEth) plus enough tokens for claims; otherwise it burns the leftover.
- `claim()`: after a successful finalize, buyers claim tokensFor(contribution).
- `graduate()`: onlyOwner; adds the reserved 15% ETH + tokens as Uniswap-V2 liquidity with the LP burned to the dead address (`0x...dEaD`).
- `abort()` / `refund()` / `refundAll()`: abort (onlyOwner, before liquidityTaken) flips to refund mode; refunds return contributions when finalized && `!softCapReached`; `refundAll` is paginated.
- `LIQUIDITY_BPS = 1500` (15%), `EARLY_FEE_BPS = 2000` (20%) are hardcoded constants.

SECURITY CONCERNS

`abort()` After `finalize()` Can Strand Buyer Refunds (HIGH - CFG04)

`finalize()` (callable by anyone once the sale ends) pays the owner 85% of the raise (`toOwner = address(this).balance - liquidityEth`) and keeps only ~15%. The owner can then call `abort()` (which only requires `!liquidityTaken`), setting `softCapReached = false` and burning the held tokens. Buyers now fall into the `refund()` path (which requires `finalized && !softCapReached`), but the contract holds only ~15% of the raise - so refunds drain that and then revert, and `claim()` also reverts (`softCap false`). Buyers of a SUCCESSFUL sale can lose ~85% of their contribution.

Recommendation: Disallow `abort()` once `finalize()` has released the owner payout (e.g. block `abort` when `finalized && softCapReached`), or escrow the owner payout until liquidity is added / the refund window closes. Make the `finalize` and `abort` state transitions mutually exclusive.

Liquidity Is Not Guaranteed; 85% Released to Owner Up Front (HIGH - CFG20)

On a successful sale, `finalize()` immediately sends 85% of the raise to the owner with no lock or vesting. Liquidity is added ONLY if the owner voluntarily calls `graduate()`; nothing in the contract forces it. An owner can take the raise and never provide liquidity, leaving

buyers holding claimable-but-illiquid tokens. When `graduate()` is called, the LP is burned to the dead address (permanently locked - a positive), but it represents only the 15% reserve. Recommendation: Provision liquidity atomically within `finalize()` (so it cannot be skipped), and/or escrow or vest the owner payout behind successful liquidity provisioning. Remove owner discretion over whether liquidity is ever created.

Systemic 15% Liquidity Ratio (MEDIUM - CFG10)

`LIQUIDITY_BPS = 1500` is a hardcoded platform default applied to EVERY sale on the pad: 85% of the raise goes to the owner and only 15% to listing liquidity. Buyers cannot choose a higher lock ratio per sale. Combined with the discretionary `graduate()` (CFG20), this is a high rug-risk ratio that contributors should understand before participating.

Recommendation: Make the liquidity ratio a per-sale, buyer-visible parameter with a sane minimum, or raise the default. Surface the ratio prominently to contributors.

20% Early-Withdraw Fee (LOW - CFG18)

`EARLY_FEE_BPS = 2000` charges a 20% fee to contributors who exit early via `withdrawEarly()`; the fee is paid to the launchpad. This is a disclosed protocol constant, but the size of the penalty should be made obvious to contributors.

Recommendation: Clearly communicate the 20% early-exit penalty in the launchpad UI before contributors commit.

Duplicated `finalize()` Logic in `graduate()` (LOW - CFG08)

`graduate()` re-implements the success branch of `finalize()` (the caps check, owner payout and burn) rather than calling a shared internal routine. Duplicated logic can diverge during maintenance and makes the state machine harder to reason about.

Recommendation: Extract the shared `finalize` logic into a single internal function called by both `finalize()` and `graduate()`.

Standalone Deployment Trusts an Arbitrary Token (LOW - CFG24)

The Presale constructor accepts an arbitrary token address. Within the factory flow the token is always a freshly deployed `TaxToken`, but a Presale deployed directly (outside the factory) could be pointed at a malicious token.

Recommendation: For the launchpad, only use presales created by the audited factory; optionally validate the token against the factory registry.

Anonymous Team - No KYC (INFORMATIONAL - CFG23)

No team member names or KYC are disclosed. This is flagged for transparency and contributes to the elevated centralization/rug-risk profile above. It also caps the score at 90 under CFG Ninja policy (Presale scores 84 on formula, below the cap).

Recommendation: Complete KYC with a recognized provider and disclose at least a team-lead identity.

POSITIVE FINDINGS

Checks-Effects-Interactions: every state-changing path (`withdrawEarly`, `finalize`, `claim`, `graduate`, `refund`, `refundAll`) updates state before external calls - no reentrancy was found (though no explicit `ReentrancyGuard` is used).

Locked Liquidity: provisioned LP is burned to the dead address = permanently locked.

Paginated Refunds: `refundAll(start, limit)` is paginated - no unbounded-loop DoS.

Bounded Buys: per-address cumulative `maxBuy` and `hardCap` bound the raise.

Supply Guaranteed: the factory's supply check reserves enough tokens for all claims plus listing.

Solidity 0.8.x: native overflow/underflow protection throughout.

AUDIT METHODOLOGY

Manual line-by-line review of the verified, flattened Solidity source (`HOODPAD.sol`) downloaded from the Robinhood Chain Blockscout contract API, mapped against the CFG01-CFG26 security framework. Automated Slither static analysis was not run for this build because the contract's compiler (`solc 0.8.34`) is newer than the `solc-select` versions pinned on the audit host; the review is a manual source analysis. On-chain state was not independently confirmed (no archive RPC/indexer for Robinhood Chain during this review).

Audit Date: September 6, 2026 | Chain: Robinhood Chain (EVM) | Auditor: CFG.NINJA | Status: PRELIMINARY

APPENDIX A: AFFECTED CODE AND SUGGESTED FIXES

CFG04 - abort() After finalize() Strands Refunds (High)

Location: Presale.finalize() -> abort() -> refund()

Affected Code:

```
function finalize() external {
...
    uint256 toOwner = address(this).balance - liquidityEth; // ~85% to owner
    if (toOwner > 0) { (bool ok,) = payable(owner).call{value: toOwner}(""); require(ok,
"eth"); }
...
}
function abort() external onlyOwner {
    require(!liquidityTaken, "lp out");
    aborted = true; finalized = true; softCapReached = false; // flips a SUCCESSFUL sale
into refund mode
    _burnLeftover(0);
...
}
```

Suggested Fix:

```
function abort() external onlyOwner {
    require(!liquidityTaken, "lp out");
    require(!(finalized && softCapReached), "already finalized success"); // block abort
after successful finalize
...
}
```

CFG20 - Liquidity Not Guaranteed / 85% Up Front (High)

Location: Presale.finalize() (owner payout) and graduate() (onlyOwner, discretionary)

Suggested Fix:

```
// Provision liquidity inside finalize() atomically instead of a separate discretionary
graduate(),
// and/or vest the 85% owner payout behind successful liquidity provisioning.
```

DISCLAIMER

This PRELIMINARY audit does not guarantee the absence of all vulnerabilities, and its findings and score are subject to change after client response and a verification (re-audit) pass. Smart-contract security is an ongoing process. This report covers the Presale contract only; the PresaleFactory and TaxToken are covered in separate reports. Users should conduct their own research before interacting with any smart contract.

I Appendix

Finding Categories

Centralization / Privilege

Centralization / Privilege findings refer to either feature logic or implementation of components that act against the nature of decentralization, such as explicit ownership or specialized access roles in combination with a mechanism to relocate funds.

Gas Optimization

Gas Optimization findings do not affect the functionality of the code but generate different, more optimal EVM opcodes resulting in a reduction in the total gas cost of a transaction.

Logical Issue

Logical Issue findings detail a fault in the logic of the linked code, such as an incorrect notion of how block.timestamp works.

Control Flow

Control Flow findings concern the access control imposed on functions, such as owner-only functions being invocable by anyone under certain circumstances.

Volatile Code

Volatile Code findings refer to segments of code that behave unexpectedly in certain edge cases that may result in a vulnerability.

Coding Style

Coding Style findings usually do not affect the generated byte-code but rather comment on how to make the codebase more legible and, as a result, easily maintainable.

Inconsistency

Inconsistency findings refer to functions that should seemingly behave similarly yet contain different code, such as a constructor assignment imposing different require statements on the input variables than a setter function.

Coding Best Practices

ERC 20 Coding Standards are a set of rules that each developer should follow to ensure the code meets a set of criteria and is readable by all developers.

Disclaimer

Scope. This report documents the results of a security assessment and smart contract audit performed by Bladepool / CFG NINJA (the "Auditor") for the project specified in this report. The assessment covers only the deliverables and code expressly listed in the engagement; any changes, integrations, or subsequent deployments are outside the scope.

No Warranty. The Auditor performs the assessment using industry-standard practices but makes no representations or warranties, express or implied, including any warranty of merchantability, fitness for a particular purpose, or non-infringement. The Auditor does not guarantee that the audited code is free of bugs, vulnerabilities, or exploitable issues.

Limitation of Liability. To the maximum extent permitted by law, the Auditor and its affiliates, officers, employees, agents, and contractors shall not be liable for any indirect, incidental, special, consequential, or punitive damages, or for any loss of profits, revenue, data, or business opportunities, arising out of or related to the assessment, even if advised of the possibility of such damages. The Auditor's aggregate liability for direct damages is limited to the fees paid for the audit engagement.

Client Responsibility. The project owner/client retains sole responsibility for all decisions and actions taken in connection with the audited code, including fixing identified issues, deploying code to any environment, and applying security mitigations. The Auditor's recommendations are advisory; implementation and verification of fixes are the client's responsibility.

Third-Party Tools and Services. The Auditor may use third-party tools, services, or automated scanners as part of the assessment. Use of such tools is without warranty; the Auditor is not responsible for defects, failures, or inaccuracies arising from third-party services.

Confidentiality and Data Security. The Auditor will treat non-public client information as confidential. However, the Auditor cannot guarantee absolute security for data transmitted over the internet or third-party platforms. Client should take reasonable precautions to protect sensitive information.

Not Financial, Legal, or Investment Advice. The assessment is a technical security review only and is not financial, legal, tax, or investment advice. Clients should consult appropriate professionals for those matters.

Ownership and Governance Notes. If ownership is transferred, held by a multisig, or controlled by a governance contract, the client should document and disclose those arrangements; the Auditor's report reflects the ownership state observed during the engagement but may not reflect subsequent changes.

Governing Law and Counsel. This engagement and any disputes arising from it shall be governed by the laws agreed in the engagement terms. Clients are advised to seek legal counsel before relying on or publishing the report.

Acceptance. By proceeding with this engagement or using this report, the client acknowledges and accepts these terms.

